

Automatic Classification of Functional Network Atoms in Mild Traumatic Brain Injury via Spatial Correlation

Derek Long
August 2026

1 Introduction

In Canada, the incidence of traumatic brain injury (TBI) is approximately 600 per 100,000 people. About 90% of these cases are classified as mild (mTBI), and one quarter of individuals experience post-concussive symptoms (PCS), which include cognitive, somatic, and emotional difficulties that hinder daily functioning, burden individuals and society, and reduce productivity (Heitger et al., 2007). mTBI is defined as an injury to the brain caused by external forces acting on the head that results in a variety of acute clinical signs and symptoms. These may include, but are not limited to, confusion lasting less than 24 hours, loss of consciousness for up to 30 minutes, or post-traumatic amnesia lasting less than 24 hours (Silverberg et al., 2023). The pathophysiology of mTBIs consists of several mechanisms:

1. **Acute Neuroinflammation** The neuroinflammatory response to mTBI is thought to play a major role in PCS. Microglial activation and infiltration occur after injury. If impact forces are sufficient to damage the blood–brain barrier, neutrophils, monocytes, and lymphocytes can enter the brain and contribute to neuronal cell death (Ghirnikar et al., 1998).
2. **Axonal Damage** In animal mTBI models, progressive axonal swelling has been shown to lead to axonal separation hours after impact (Mata et al., 1986). The likely underlying mechanism is rotational acceleration and deceleration of the brain, which stresses neurons due to their rigidity at the grey and grey–white matter junctions. Membrane disruption and depolarization caused by axonal stretching led to Ca^{2+} influx and mitochondrial swelling, ultimately resulting in constriction and secondary axotomy, the physical severing of the axon in the days following injury (Maxwell et al., 1995).
3. **Reduced Cerebral Blood Flow (CBF) and Altered Metabolism** Vascular injury following mTBI triggers an inflammatory response in the brain parenchyma, affecting neuronal function and metabolism. Trauma-induced intracellular increases in Ca^{2+} stimulate the expression and redistribution of endothelin, a vasoconstrictor that reduces CBF and cerebrovascular reactivity (Rafols et al., 2007).

Diffusion tensor imaging has become the most widely used modality in the mTBI literature. However, in previous studies, DTI produced mixed results when quantifying diffusion biomarkers related to mTBI (Lunkova et al., 2025; Lunkova et al., 2021). To overcome these limitations, our lab employs three different imaging techniques. Sparsity-based Analysis of Reliable K-hubness (SPARK), one of these techniques, will be used to examine how functional networks and connectivity “hubs” reorganize after mTBI (Lee et al., 2016). However, SPARK requires manual classification of the “atoms”, temporal or spatial features of a functional network, which is a tedious and error-prone process. This project therefore aims to automatically classify SPARK atoms to increase the reproducibility, precision, recall, and accuracy in the context of mTBI to allow for reliable results across all time points of a longitudinal study. The final goal is to build an automated spatial-correlation classifier using complementary brain atlases that can match manual atom classification accuracy to eliminate the labor bottleneck.

2 Background

2.1 Sparsity-based Analysis of Reliable K-hubness (SPARK)

Brain activity is organized in functional networks consisting of spatially distributed brain regions whose blood-oxygen-level-dependent (BOLD) signals fluctuate synchronously over time, reflecting coordinated cognitive, motor, sensory, and affective processes (Power et al., 2011). Several studies have shown that mTBI leads to neural plasticity causing a change in the connectivity of functional brain networks (Dimitriadis et al., 2015; España-Irla et al., 2025; Healey et al., 2022). For example, Kim et al. found that functional connectivity between the default mode network and dorsal/ventral network is significantly reduced in mTBI patients. Our group hypothesizes that “hub-like” brain regions with global functional network centrality may be the key to understanding changes in functional connectivity following mTBI. To investigate this possibility, SPARK is used to identify functional networks and the overlapping regions between them, known as “connector hubs”, longitudinally in a cohort of mTBI patients and matched healthy controls (Lee et al. 2016).

Traditionally, the detection of these hubs by Independent Component Analysis, hierarchical clustering, and graph theory presents many limitations (Lee et al., 2016):

1. Multi-collinearity where the highly correlated nodes of brain networks make detection of individual networks difficult.
2. Individual networks are identified from non-overlapping regions.
3. These techniques are biased by network size.
4. They do not identify the number and type of network involved in hubs.

5. They produce varied results based on individual differences in physiology

However, SPARK resolves each of these limitations by respectively

1. Handling each node at the voxel level.
2. Using a biologically plausible sparsity assumption to identify networks.
3. Being impartial to network size.
4. Clearly identifying the number and type of network involved in hubs.
5. Showing statistically reproducible results.

SPARK uses preprocessed resting-state fMRI (rs-fMRI). The resulting whole-brain fMRI data, denoted by Y , are used to estimate the parameters of a sparse generalized linear model (GLM). A bootstrap resampling strategy that preserves the temporal structure of the fMRI signal is used to obtain B resampled datasets, denoted by Y_b , for $b = 1, \dots, B$.

For each Y_b , a dictionary of temporal features Ω_b and its corresponding sparse coefficient matrix X_b are estimated by solving the optimization problem

$$\min_{\Omega_b, X_b} \|Y_b - \Omega_b X_b\|_F^2$$

subject to

$$\|x_{b,i}\|_0 \leq k_i,$$

where

- Y_b denotes the b th resampled preprocessed fMRI dataset,
- Ω_b denotes the temporal dictionary,
- X_b denotes the sparse spatial coefficient matrix,
- $x_{b,i}$ denotes the coefficient vector associated with voxel i in bootstrap sample b ,
- k_i denotes the voxel-specific sparsity constraint, and
- $\|\cdot\|_F$ denotes the Frobenius norm (Herzog et al., 2023).

The resulting outputs are then collected and processed using a spatial clustering algorithm to estimate $N_{\text{sig+noise}}$, which represents the number of detected atoms, including both reliable signal components and noise components, in each dictionary. The final value N_{sig} is obtained after statistically unreliable elements are removed (*Figure 1*). Noisy components are manually removed after visual inspection of each atom.

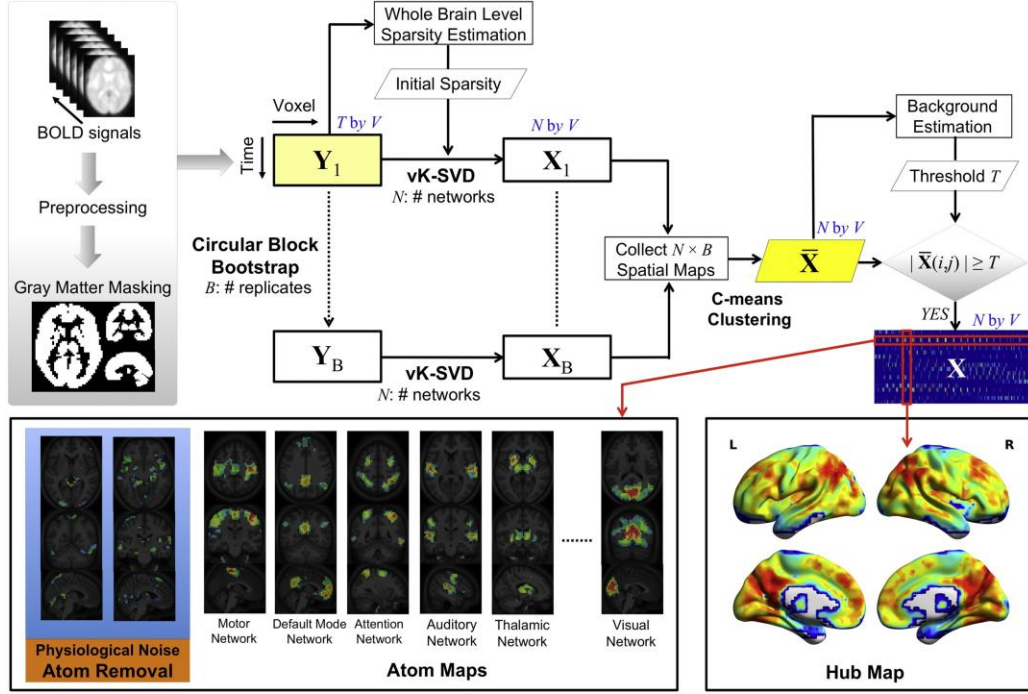


Figure 1. Summary of SPARK Pipeline. Adapted from Lee et al. 2016.

3 Methods

We used preprocessed rs-fMRI images of three healthy controls from a previous cross-sectional imaging study (HC043, HC044, and HC048) and ran them through SPARK to acquire the atoms for each subject as Nifti image files (Dascal, 2023). Acquisition of N_{sig} requires manual identification and rejection of noisy atoms, so a custom spatial-correlation classifier implemented using Nilearn automates the process by classifying each atom into its corresponding functional network. We chose 15 commonly encountered functional brain networks based on previous literature in the field, as follows:

1. Auditory Network
2. Cerebellum
3. Dorsal Attention Network (DAN)
4. Default Mode Network (DMN)
5. Executive Control Network
6. Precuneus
7. Hippocampus
8. Salience Network

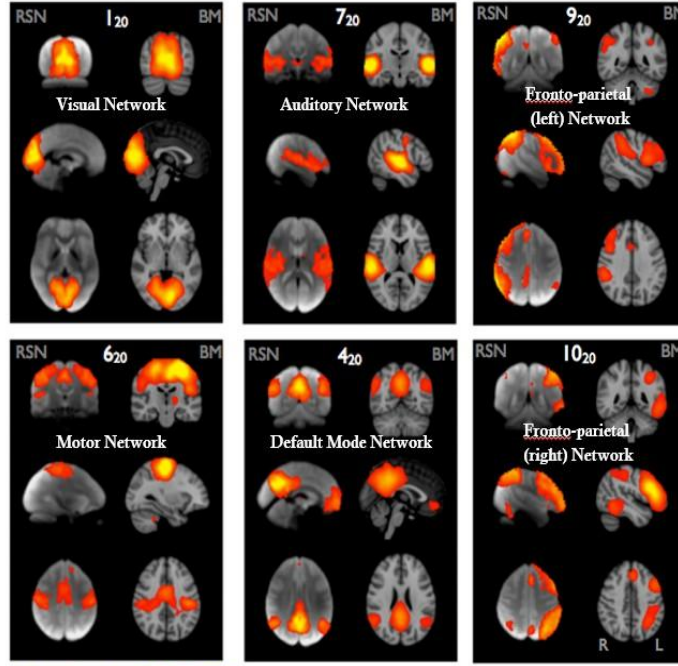
9. Sensorimotor Network
10. Subcortical Network
11. Temporal Lobe
12. Visual Lateral Network
13. Visual Medial Network
14. Visual Occipital Network
15. Visuospatial Network

There is also a "noise label" for any atom whose BOLD fMRI activity does not fit any of the above networks because SPARK occasionally wrongly identifies network atoms comprised of patterns corresponding to biological noise, such as cerebrospinal fluid, major blood vessels, and other physiological activity.

3.1 Functional Network Classification via Spatial Correlation

The classifier uses three complementary brain atlases because each provides a different representation of the brain regions and functional networks considered in this study.

The Smith Atlas provides canonical resting-state functional network maps (Collins et al., 1998; Laird et al., 2011; Smith et al., 2009; Tzourio-Mazoyer et al. 2002). Additional anatomical templates from the Harvard-Oxford and AAL atlases, both at 2-mm resolution, were used for regions not directly represented by the selected Smith functional maps, including smaller cortical subregions and the cerebellum (Desikan et al. 2006; Frazier et al. 2005; Goldstein et al. 2007; Makris et al. 2006). Executive control and sensorimotor templates derived from this atlas combination were excluded from the final classifier because preliminary testing showed that their inclusion reduced classification precision and recall (*Figure 2*).



Smith et al., PNAS 2009

Figure 2. Smith atlas resting-state network maps. Adapted from Smith et al. (2009).

The Schaefer atlas provides cortical functional parcellations (Schaefer et al. 2018). We used the 400-region atlas with the 17-network parcellation at a 2-mm resolution. Visuospatial, subcortical, temporal lobe, and cerebellar categories were omitted because these categories were either not directly represented by the cortical parcellation or their inclusion reduced classification performance.

MIST-122 was used for finer whole-brain functional parcellation that includes cortical, subcortical, and cerebellar structures (Urchs et al. 2017). The MIST-122 parcellation and its associated parcel metadata were regrouped into the 15 anatomical and functional categories targeted by the classifier by assigning individual labeled parcels to corresponding network masks. The MIST-derived hippocampal template was excluded from the final classifier after preliminary testing indicated that it reduced classification performance.

These atlas-specific exclusions were determined empirically during classifier development and therefore represent a potential source of selection bias (see Section 5.3).

Each input atom is represented as a three-dimensional spatial map and flattened into a one-dimensional vector. For each target network, the corresponding atlas template is first resampled to the spatial grid of the subject atom, the Montreal Neurological Institute 152 nonlinear 2009c asymmetrical template (Fonov et al., 2009) and then flattened in the same manner. Spatial similarity is quantified using Pearson’s product-moment correlation computed with *np.corrcoef*. The absolute value of the correlation coefficient is used as a

simplification so that the score for each network reflects the strength of spatial correspondence.

For each target label, the final similarity score is calculated as the mean of the available atlas-specific correlation scores. Atlases that do not contain a template for a particular category are omitted from that category’s average. The labels are subsequently ranked in descending order of similarity, with the highest-scoring label representing the predicted classification.

A noise score threshold of 0.18 was selected empirically to maximize the weighted classification F1 score for all three controls. Atoms whose highest similarity score falls below this threshold are classified as noise for later manual removal. Because the same subjects were used both to select this threshold and to evaluate the classifier, the reported results should be interpreted as preliminary rather than as an independent validation.

To reduce repeated computation, memoization is used to cache resampled atlas templates. The cache key contains the subject atom’s spatial dimensions, affine transformation, atlas name, and network label, allowing a previously resampled template to be reused when an equivalent spatial grid is encountered. Overall, the time complexity of the algorithm is polynomial with $O(ALGD)$ where

- A denotes the number of atoms,
- L denotes the number of target labels,
- G denotes the number of atlas templates evaluated per label, and
- D denotes the number of voxels in each flattened spatial map.

Since the G and D are fixed for a given configuration, the classification cost scales approximately as $O(AL)$ with respect to the number of atoms and target labels.

4 Results

The classifier achieved an overall accuracy of 81% on HC043 and HC048, and 85% on HC044. Macro- and weighted-average F1 scores were 84% and 81% for HC043, 88% and 85% for HC044, and 81% and 79% for HC048, respectively. These metrics represent substantial improvements from the first version of the classifier that used only the Smith Atlas and showed accuracy, macro and weighted average F1 scores of respectively 27%, 20%, and 24% (*Figure 3*). Nonetheless, the classifier currently struggles to classify regions associated with the cerebellum, DMN, executive control network, precuneus, and temporal lobe, and it cannot classify DAN (*Figures 4-6*).

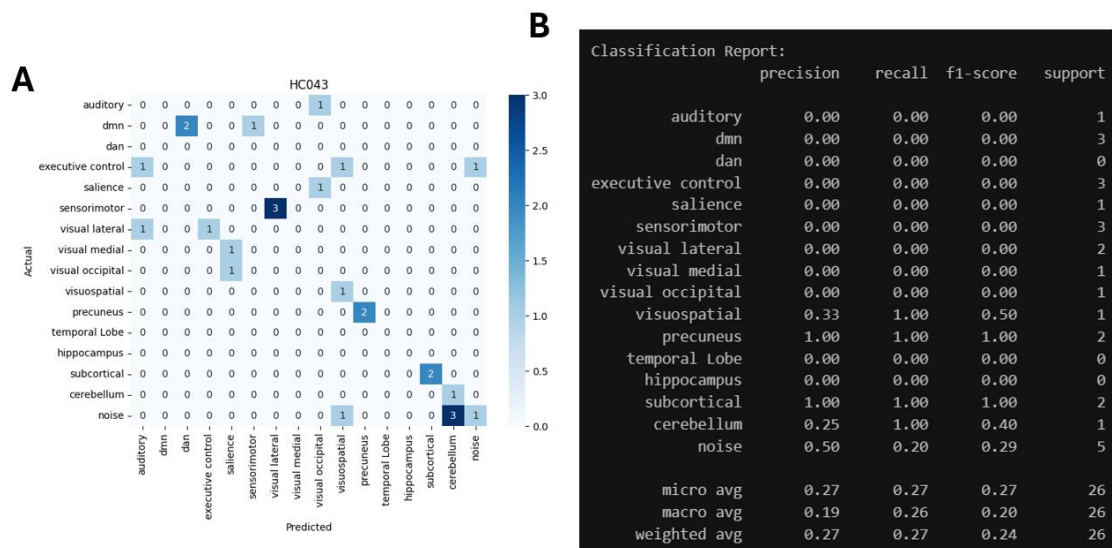


Figure 3. Baseline classification results for HC043 on the first classifier prototype. A) Confusion matrix; B) Performance metrics for the classifier.

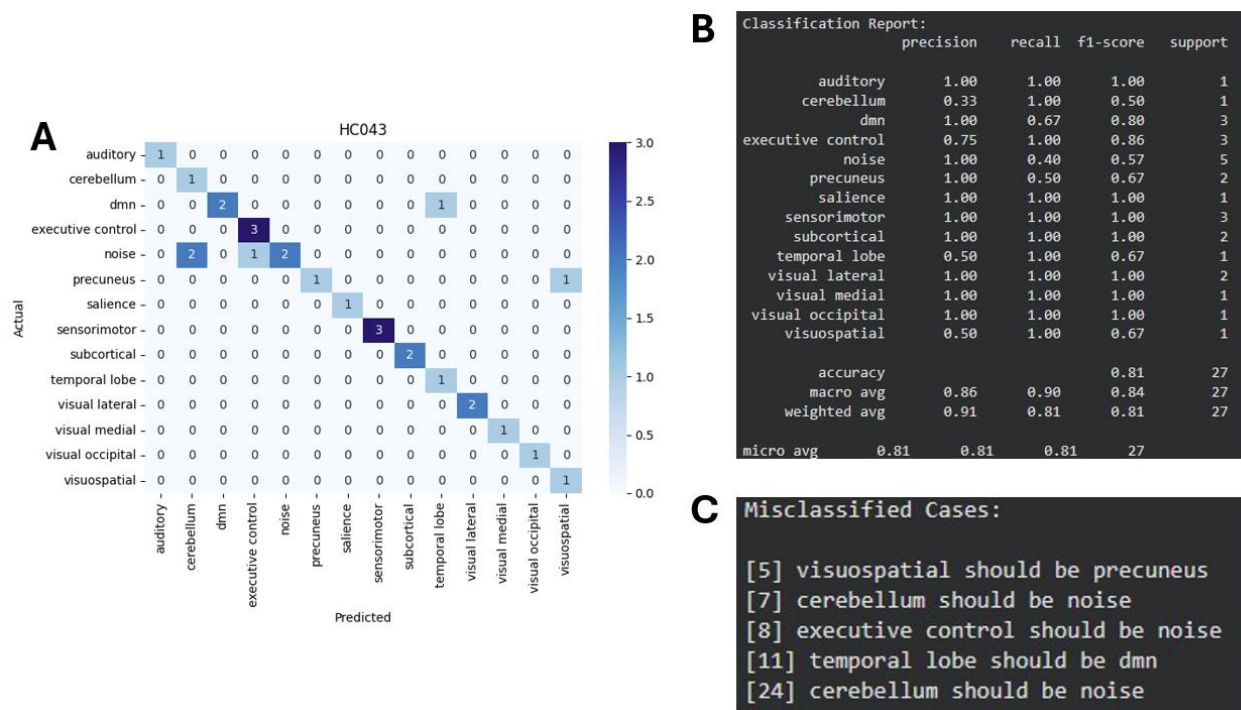


Figure 4. Final classification results for HC043. A) Confusion matrix; B) Performance metrics for the classifier; C) Details of misclassified cases for this output sorted by atom number.

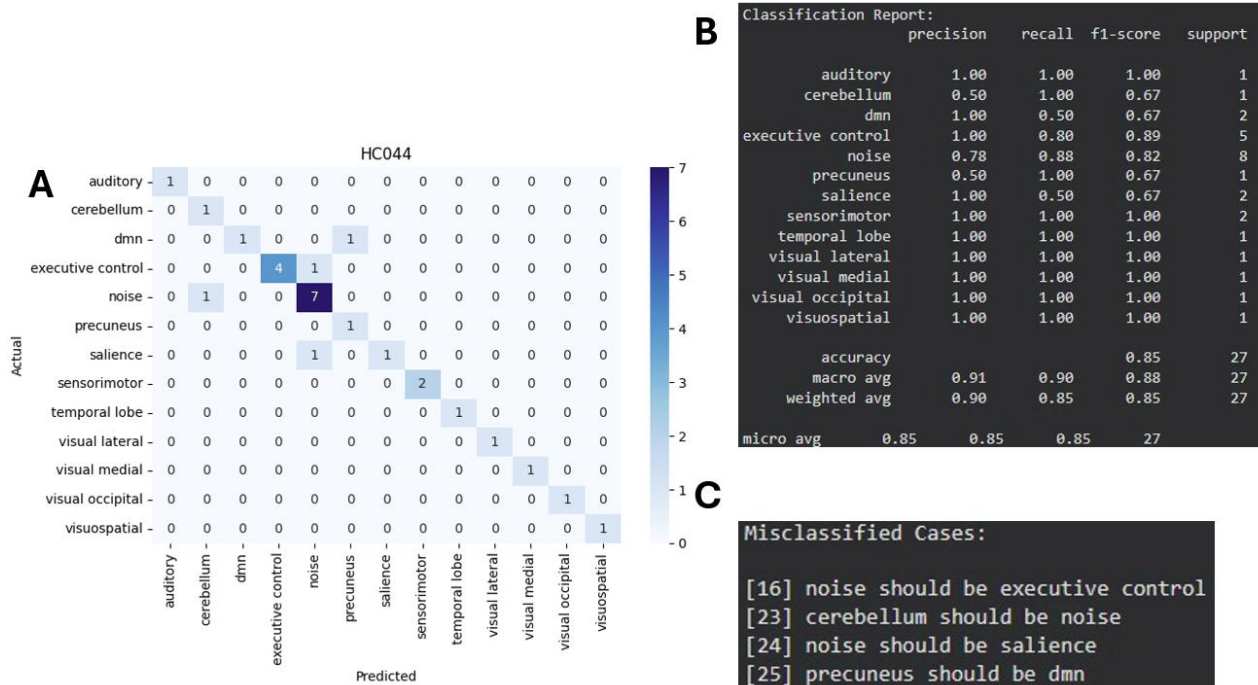


Figure 5. Final classification results for HC044. A) Confusion matrix; B) Performance metrics for the classifier; C) Details of misclassified cases for this output sorted by atom number.

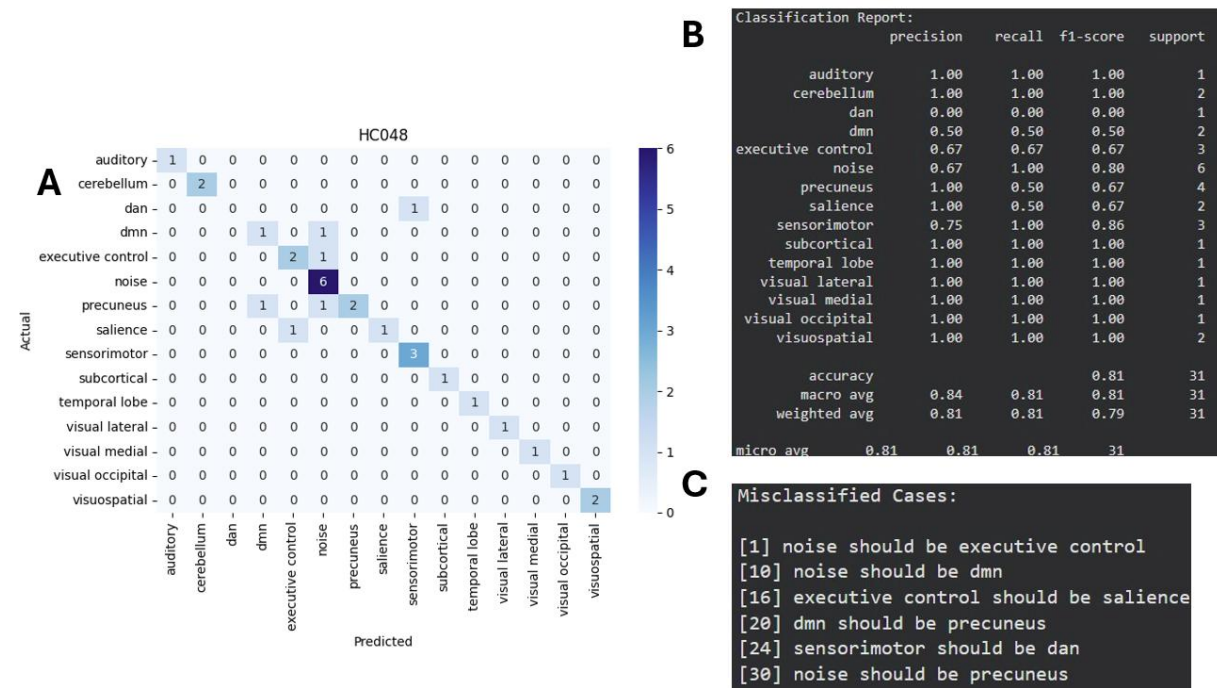


Figure 6. Final classification results for HC048. A) Confusion matrix; B) Performance metrics for the classifier; C) Details of misclassified cases for this output sorted by atom number.

5 Discussion

In the full scope of this research, our group plans to assess patients with acute mTBI using resting-state fMRI longitudinally. For comparison, two matched control groups will be recruited: a neurologically intact control group and an orthopedic group with patients having sustained a non-surgical injury (orthopedic injury excluding cervical spine with no comorbid TBI). SPARK data will be combined with the outputs of other advanced neuroimaging modalities to generate a combined risk score for PCS outcome following mTBI. In preparation for completed recruitment and full-scope analysis, a bottleneck was identified in the SPARK pipeline, which highlighted the need for an automatic atom classifier. This classifier provides the potential not only to overcome temporal limitations in the SPARK pipeline, where many man hours are required to accurately identify networks corresponding to each atom, but further has the promise of increasing reproducibility of atom labelling across timepoints, which is especially relevant to this longitudinal study.

I chose a deterministic spatial-correlation approach because only three healthy-control datasets were available, and the distribution of network classes was highly imbalanced. Under these conditions, training a supervised machine-learning model such as a convolutional neural network (CNN) would not be appropriate because substantially more labeled examples would be required to reliably estimate model parameters and evaluate generalization. The spatial-correlation approach also offers greater interpretability for the present application. Rather than producing only a predicted class, the classifier retains the similarity score associated with each candidate's functional network, allowing individual predictions and competing classifications to be inspected directly.

5.1 Computational Costs

The brain atlases used in this study provide high-quality, previously validated spatial templates without requiring the classifier to learn spatial representations from a large training dataset. In contrast, training a CNN for 15 classes would require a substantially larger and more balanced labeled dataset, as well as greater computational resources. Though a significantly larger dataset will be available for training in the future, neither was available in the present study, providing a significant limitation. The spatial-correlation classifier is comparatively inexpensive to evaluate, and repeated template resampling was further reduced through memoization.

5.2 Reproducibility

The use of deterministic brain atlases for spatial correlation resulted in consistent accuracies on the classification of the three different healthy controls. In contrast, a linear model with batch normalization under stochastic gradient descent might generate different results due to random data variation (Han, 2025). Since the full scope of the research involves evaluating patients longitudinally, it will require accurate classification of the atoms across all timepoints to accurately quantify the changes in mTBI patient data.

5.3 Limitations

A major challenge for the classifier is the heterogeneity of the target classes representing different types of regions of interest (ROIs). Functional networks include DMN, DAN, salience, executive control, sensorimotor, and auditory. The precuneus, hippocampus, temporal lobe, subcortical, and cerebellum are anatomical regions. Visual medial, visual lateral, visual occipital, and visuospatial are all hybrid/custom visual divisions. Therefore, many of these regions overlap and might cause the classifier to be confused. For example, both the hippocampus and precuneus overlap with the DMN, the DAN overlaps with the visuospatial network, and the visual occipital network overlaps with the visual medial and lateral networks (*Figure 7*). Since the MIST template for DAN was omitted, the other parcellations failed to correctly identify atom 24 in HC048 which led to the confusion with sensorimotor. The classifier is also sensitive to noise and would classify noisy physiological activity as regions that share a high similarity with actual networks (*Figure 8*). These issues need to be addressed before running the classifier on mTBI patients since in their case, SPARK can produce atoms with noisier or different spatial signatures to healthy controls.

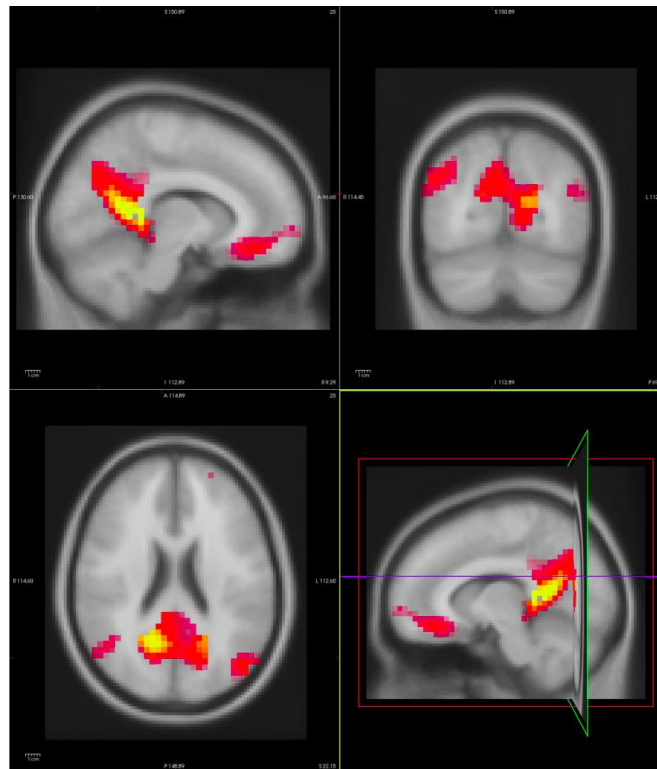


Figure 7. Atom 13 from HC043 is labeled as precuneus HC043, but it exhibits activity from the hippocampus and the DMN.

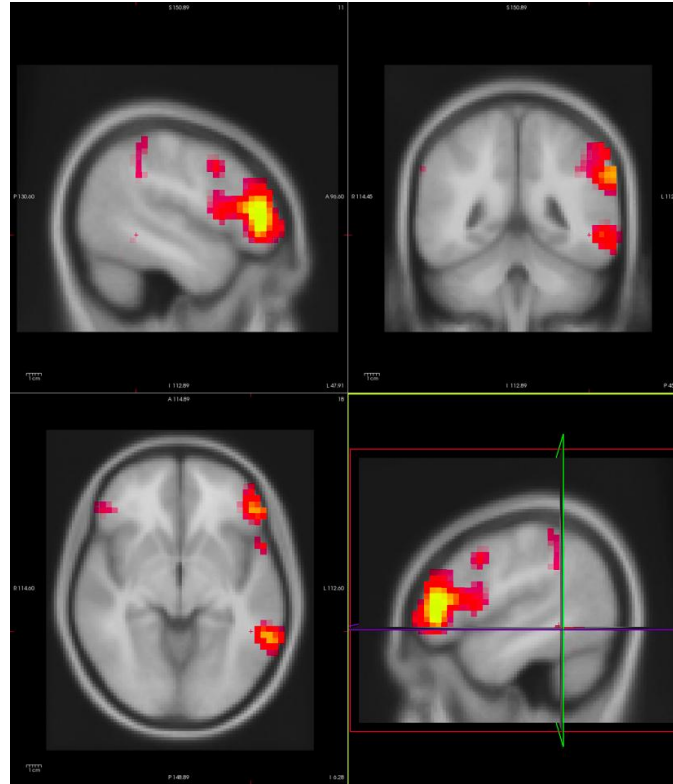


Figure 8. Atom 8 from HC043 is wrongly classified as executive control since it is noise from the cerebrospinal fluid in the subarachnoid space.

Moreover, the labeling of the dataset for testing was also done manually, which introduces the potential for biased labelling due to human error. Upon visualization of the atoms on brain maps, the classification of the atoms produced by SPARK could be ambiguous since they present noisy patterns that look like actual functional networks when they could just be cerebrospinal fluid or blood vessels inside the brain. The approach used to quantify noise is crude, and the noise threshold might overestimate noise for some patients if it is too high or incorrectly classify noisy atoms if it is too low (*Figure 9*). Manually assigned reference labels introduce potential bias into the evaluation, since measured classifier performance depends on the accuracy and consistency of the human annotations.

An additional limitation is that several design choices were optimized using the same three healthy controls that were later used to report on classifier performance. These choices included the empirical noise threshold and the exclusion of selected atlas-derived templates that reduced the classification accuracy. Consequently, the reported accuracies may overestimate performance on unseen subjects. A more rigorous evaluation would separate classifier development and validation by determining thresholds and atlas weights on a training dataset and evaluating the finalized classifier on an independent test dataset. This will be pursued in the future when a complete cohort has been recruited.

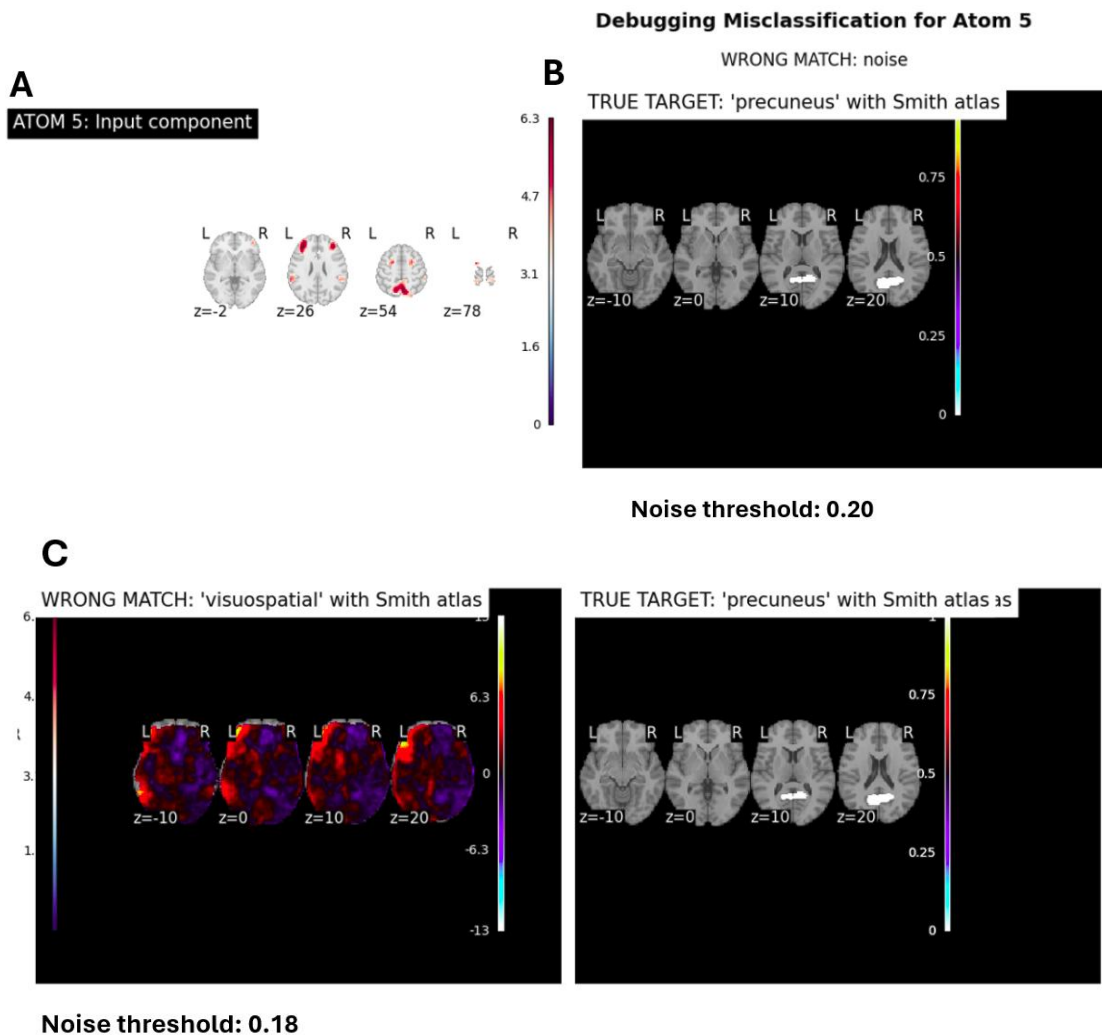


Figure 9. Misclassification of atom 5 due to noise threshold. A) Visualization of atom 5; B) Wrongly identified as noisy if the noise threshold is increased to 0.20; C) Wrongly classified as "visuospatial" instead of "precuneus" because of the noisy L/R activity if the noise threshold is below 0.18.

Furthermore, averaging similarity scores across the available atlas templates may bias the final classification toward atlases whose spatial representation better matches a particular network. For example, the Smith parcellation might cover a region that is not present in finer parcellations like Schaefer or MIST, like the medial-frontal cortex, anterior cingulate cortex (ACC), paracingulate, and bilateral anterior insula, leading to bias towards that atlas for networks containing these regions (Smith et al., 2009).

5.4 Future Directions

Future work should first evaluate the classifier on a larger and more balanced dataset containing independently labeled SPARK atoms. The complete dataset should be randomly divided into subject-level train/test sets so that parameters such as the noise threshold, atlas selection, and atlas weights can be optimized using machine learning approaches without biasing the final performance estimate.

To address the confusion of certain networks like DMN and visual networks, classification could also be performed at a finer spatial level, for example by distinguishing anterior and posterior subdivisions of the DMN, to reduce ambiguity between overlapping anatomical and functional categories. Rather than assigning equal weight to every atlas, atlas-specific weights could be estimated to account for differences in spatial coverage and suitability for individual networks.

Finally, noise detection could be improved by incorporating dedicated templates for cerebrospinal fluid, ventricles, major blood vessels, and other physiological or imaging artifacts. Additional pre-processing to remove physiological nuisance variables may also aid in reducing noise during the classification step.

6 Conclusion

This project developed an automated spatial-correlation classifier to address the manual atom-classification bottleneck in the SPARK pipeline. By combining complementary templates derived from the Smith, Schaefer, and MIST atlases, the classifier achieved overall accuracies of 81-85% across three healthy controls, demonstrating the feasibility of automatically identifying many of the functional and anatomical patterns detected by SPARK. The current results remain preliminary because of the small and highly imbalanced evaluation dataset, overlap between several target categories, uncertainty in manually assigned reference labels, and the use of the same subjects for classifier development and evaluation. Despite these limitations, the classifier provides an interpretable and computationally efficient approach for reducing manual processing within the SPARK pipeline and establishes a foundation for future application of SPARK to the study of functional network reorganization following mTBI.

7 Data Availability

The Python notebook for the atom classifier can be found here:

https://colab.research.google.com/drive/1P6O222bmlqURKvus3Afp8_MFMo_9Cnug#scrollTo=f5012977

8 Statement on the use of LLMs

LLMs were used to assist with code development, including code structuring and debugging, guidance on the implementation of spatial correlation, and the parcellation of brain atlases. They were also used to assist with reformulating and shortening sentences throughout the report.

9 References

Collins, D. L., Zijdenbos, A. P., Kollokian, V., Sled, J. G., Kabani, N. J., Holmes, C. J., & Evans, A. C. (1998). Design and construction of a realistic digital brain phantom. *IEEE Transactions on Medical Imaging*, 17(3), 463–468. <https://doi.org/10.1109/42.712135>

Dascal, A. (2023). *Whole brain profile of connector hub alteration patterns in focal epilepsy when compared to healthy controls* [Master's thesis, Concordia University]. Spectrum Research Repository. <https://spectrum.library.concordia.ca/id/eprint/992727/>

Desikan, R. S., Ségonne, F., Fischl, B., Quinn, B. T., Dickerson, B. C., Blacker, D., Buckner, R. L., Dale, A. M., Maguire, R. P., Hyman, B. T., et al. (2006). An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. *NeuroImage*, 31(3), 968–980.

Dimitriadis, S. I., Zouridakis, G., Rezaie, R., Babajani-Feremi, A., & Papanicolaou, A. C. (2015). Functional connectivity changes detected with magnetoencephalography after mild traumatic brain injury. *NeuroImage: Clinical*, 9, 519–531. <https://doi.org/10.1016/j.nicl.2015.09.011>

España-Irla, G., Tinney, E. M., Ai, M., Nwakamma, M., & Morris, T. P. (2025). Functional connectivity patterns following mild traumatic brain injury and the association with longitudinal cognitive function. *Human Brain Mapping*, 46, e70237. <https://doi.org/10.1002/hbm.70237>

Fonov, V. S., Evans, A. C., McKinstry, R. C., Almli, C. R., & Collins, D. L. (2009). Unbiased nonlinear average age-appropriate brain templates from birth to adulthood. *NeuroImage*, 47(Suppl. 1), S102. [https://doi.org/10.1016/S1053-8119\(09\)70884-5](https://doi.org/10.1016/S1053-8119(09)70884-5)

Frazier, J. A., Chiu, S., Breeze, J. L., Makris, N., Lange, N., Kennedy, D. N., Herbert, M. R., Bent, E. K., Koneru, V. K., Dieterich, M. E., et al. (2005). Structural brain magnetic resonance imaging of limbic and thalamic volumes in pediatric bipolar disorder. *American Journal of Psychiatry*, 162(7), 1256–1265.

Ghirnikar, R. S., Lee, Y. L., & Eng, L. F. (1998). Inflammation in traumatic brain injury: Role of cytokines and chemokines. *Neurochemical Research*, 23(3), 329–340.

Goldstein, J. M., Seidman, L. J., Makris, N., Ahern, T., O'Brien, L. M., Caviness, V. S., Jr., Kennedy, D. N., Faraone, S. V., & Tsuang, M. T. (2007). Hypothalamic abnormalities in schizophrenia: Sex effects and genetic vulnerability. *Biological Psychiatry*, 61(8), 935–945.

Han, H. (2025). Challenges of reproducible AI in biomedical data science. *BMC Medical Genomics*, 18(Suppl. 1), 8. <https://doi.org/10.1186/s12920-024-02072-6>

Healey, K., Fang, Z., Smith, A., Zemek, R., & Ledoux, A.-A. (2022). Adolescents with a concussion have altered brain network functional connectivity one month following injury when compared to adolescents with orthopedic injuries. *NeuroImage: Clinical*, 36, 103211. <https://doi.org/10.1016/j.nicl.2022.103211>

Heitger, M. H., Jones, R. D., Frampton, C. M., et al. (2007). Recovery in the first year after mild head injury: Divergence of symptom status and self-perceived quality of life. *Journal of Rehabilitation Medicine*, 39, 612–621.

Herzog, R., Köhne, F., Kreis, L., & Schiela, A. (2023). Frobenius-type norms and inner products of matrices and linear maps with applications to neural network training [Preprint]. *arXiv*. <https://doi.org/10.48550/arXiv.2311.15419>

Kim, E., Seo, H. G., Yoo, R.-E., & Oh, B.-M. (2026). Time course of functional and structural brain network changes after mild traumatic brain injury. *Brain Communications*, 8(2), fcag072. <https://doi.org/10.1093/braincomms/fcag072>

Laird, A. R., Fox, P. M., Eickhoff, S. B., Turner, J. A., Ray, K. L., McKay, D. R., Glahn, D. C., Beckmann, C. F., Smith, S. M., & Fox, P. T. (2011). Behavioral interpretations of intrinsic connectivity networks. *Journal of Cognitive Neuroscience*, 23(12), 4022–4037. https://doi.org/10.1162/jocn_a_00077

Lee, K., Lina, J. M., Gotman, J., & Grova, C. (2016). SPARK: Sparsity-based analysis of reliable k-hubness and overlapping network structure in brain functional connectivity. *NeuroImage*, 134, 434–449. <https://doi.org/10.1016/j.neuroimage.2016.03.049>

Lunkova, E., Amir, J., Chen, J.-K., McCabe, S., Saluja, R. S., & Ptito, A. (2025). Multimodal approach in the identification of biomarkers of mild traumatic brain injury: Resting state fMRI, ASL, and SWI. *Clinical Neuroimaging*, 2(1), e70026. <https://doi.org/10.1002/neo2.70026>

Lunkova, E., Guberman, G. I., Ptito, A., & Saluja, R. S. (2021). Noninvasive magnetic resonance imaging techniques in mild traumatic brain injury research and diagnosis. *Human Brain Mapping*, 42(16), 5477–5494. <https://doi.org/10.1002/hbm.25630>

Makris, N., Goldstein, J. M., Kennedy, D., Hodge, S. M., Caviness, V. S., Faraone, S. V., Tsuang, M. T., & Seidman, L. J. (2006). Decreased volume of left and total anterior insular lobule in schizophrenia. *Schizophrenia Research*, 83(2–3), 155–171. <https://doi.org/10.1016/j.schres.2005.11.020>

Mata, M., Staple, J., & Fink, D. J. (1986). Changes in intra-axonal calcium distribution following nerve crush. *Journal of Neurobiology*, 17, 449–467.

Maxwell, W. L., McCreath, B. J., Graham, D. I., & Gennarelli, T. A. (1995). Cytochemical evidence for redistribution of membrane pump calcium-ATPase and ecto-Ca-ATPase activity, and calcium influx in myelinated nerve fibres of the optic nerve after stretch injury. *Journal of Neurocytology*, 24, 925–942.

Power, J. D., Cohen, A. L., Nelson, S. M., et al. (2011). Functional network organization of the human brain. *Neuron*, 72(4), 665–678. <https://doi.org/10.1016/j.neuron.2011.09.006>

Rafols, J. A., Kreipke, C. W., & Petrov, T. (2007). Alterations in cerebral cortex microvessels and the microcirculation in a rat model of traumatic brain injury: A correlative EM and laser Doppler flowmetry study. *Neurological Research*, 29(4), 339–347.

Schaefer, A., Kong, R., Gordon, E. M., Laumann, T. O., Zuo, X.-N., Holmes, A. J., Eickhoff, S. B., & Yeo, B. T. T. (2018). Local-global parcellation of the human cerebral cortex from intrinsic functional connectivity MRI. *Cerebral Cortex*, 29, 3095–3114.

Silverberg, N. D., et al. (2023). The American Congress of Rehabilitation Medicine diagnostic criteria for mild traumatic brain injury. *Archives of Physical Medicine and Rehabilitation*, 104(8), 1343–1355. <https://doi.org/10.1016/j.apmr.2023.03.036>

Smith, S. M., Fox, P. T., Miller, K. L., Glahn, D. C., Fox, P. M., Mackay, C. E., Filippini, N., Watkins, K. E., Toro, R., Laird, A. R., & Beckmann, C. F. (2009). Correspondence of the brain's functional architecture during activation and rest. *Proceedings of the National Academy of Sciences*, 106(31), 13040–13045. <https://doi.org/10.1073/pnas.0905267106>

Tzourio-Mazoyer, N., Landeau, B., Papathanassiou, D., Crivello, F., Etard, O., Delcroix, N., Mazoyer, B., & Joliot, M. (2002). Automated anatomical labeling of activations in SPM using a macroscopic anatomical parcellation of the MNI MRI single-subject brain. *NeuroImage*, 15(1), 273–289. <https://doi.org/10.1006/nimg.2001.0978>

Urchs, M., Armoza, J., Benhajali, Y., et al. (2017). *MIST: A multi-resolution parcellation of functional networks* [Data set]. Figshare. <https://doi.org/10.6084/m9.figshare.5633638>